

Asm 3 (All nums 4 bit)

(MAJID)

Q1: Adder-Subtractor if A is divisible by 4 and B is greater than 7 then do(A-B) else do(A+B)

Cin = 1 % if $(A \% 4 = 0) \cdot \text{AND} (B > 7)$

need to find eqn for both cases.

as All nums 4 bits: $A = xyzw = A_3 A_2 A_1 A_0$ $B = xyzw = B_3 B_2 B_1 B_0$

	x	y	z	w
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

$A \% 4 = 0$	$B > 7$
1	0
0	0
0	0
0	0
1	0
0	0
0	0
0	0
1	1
0	1
0	1
0	1
1	1
0	1
0	1
0	1

$A \% 4 = 0$

zw	00	01	11	10
xy 00	1	0	0	0
01	1	0	0	0
11	1	0	0	0
10	1	0	0	0

$$(A \% 4) = \overline{zw} = \overline{A_1 A_0}$$

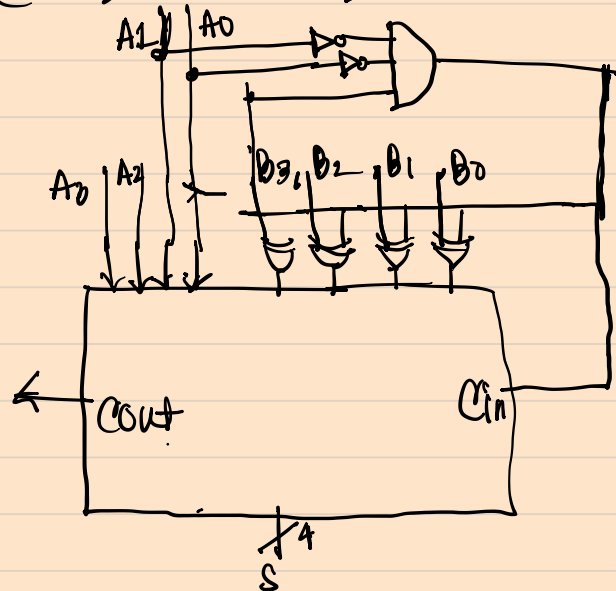
$B > 7$

zw	00	01	11	10
xy 00				
01				
11	1	1	1	1
10	1	1	1	1

$$B > 7 = x = B_3$$

$C_{in} = 1$ if $(A \% 4 = 0)$ AND $(B \% 4 = 0)$

$$C_{in} = (\overline{A_1} \overline{A_0}) \text{ AND } (B_3) = \overline{A_1} \cdot \overline{A_0} \cdot B_3$$



Q2: if $A \% 3 = 0$ AND $B \% 3 = 0$ then $A - B$ else $A + B$
 $A = A_3 A_2 A_1 A_0$ $B = B_3 B_2 B_1 B_0$

$C_{in} = 1$ if $(A \% 3 = 0)$ AND $(B \% 3 = 0)$

Make Truth table like before if you need
 for $(0, 3, 6, 9, 12, 15)$ you will get $(A \% 3 = 0)$
 and $(B \% 3 = 0)$

$A \% 3 = 0$

A \ B	00	01	10	11
00	1		1	
01				1
11	1		1	
10		1		

Nothing to take common

$$(A \% 3 = 0)$$

$$= \overline{A_3} \overline{A_2} \overline{A_1} \overline{A_0} + \overline{A_3} \overline{A_2} A_1 A_0 + \overline{A_3} A_2 \overline{A_1} \overline{A_0} + \overline{A_3} A_2 A_1 A_0 + \overline{A_3} A_2 \overline{A_1} A_0 + \overline{A_3} A_2 A_1 A_0$$

$$(A \% 3 = 0) = \overline{A_3} \overline{A_2} (A_1 \oplus A_0) + A_3 A_0 (A_1 \oplus A_2) + A_2 \overline{A_0} (A_3 \oplus A_1) \quad \text{--- (1)}$$

Similarly:

$$(B \% 3 = 0) = \overline{B_3} \overline{B_2} (B_1 \oplus B_0) + B_3 B_0 (B_1 \oplus B_2) + B_2 \overline{B_0} (B_3 \oplus B_1) \quad \text{--- (2)}$$

directly can make this



$Cin = 1$ if $(A \% 3 = 0)$ AND $(B \% 3 = 0)$

$Cin = ① + ②$

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